

CO1 Home Assignment

23AAS3205R AUTONOMOUS VEHICLE ECU DESIGN WITH AUTOSAR

1. Illustrate the evaluation of protocols used in automotive system and how to select a protocol for new features. Compare the LIN and CAN communication protocols in terms of architecture, data rate, and cost. Justify why LIN is more suitable than CAN for body electronics applications.
2. List out the core Unified Diagnostic Services (UDS) of ISO 14229:2020 and discuss their operation in automotive ECUs. Analyze the ISO 14229:2020 Unified Diagnostic Services (UDS) standard used in road vehicles. Examine the UDS protocol architecture and diagnostic service categories. Discuss the role and significance of UDS in modern EV and autonomous vehicles.
3. Analyze the Diagnostic Communication Manager (DCM) in AUTOSAR and its role in automotive diagnostics. Examine the different sub modules of DCM and their specific functions. Discuss the importance of DCM sub modules in ensuring reliable ECU communication and diagnostics.
4. Analyze the XCP data organization in automotive ECUs used for real-time measurement and calibration. Examine how measured variables (DAQ), calibration parameters (CAL), and stimulus data (STIM) are structured in memory.
5. Analyze the XCP protocol architecture used for measurement and calibration in automotive ECUs. Explain the layered communication structure, and the role of DAQ (Data Acquisition) and STIM (Stimulus) in real-time measurement. Compare and analyze the use of XCP, CCP, and UDS protocols for measurement and calibration in EVs and AVs.